

Reciprocal Lerch Products

Proposed ProveIt manuscript insertion

Companion to *Reciprocal Lerch Convolutions*

10 October 2026

1 Reciprocal Lerch products and entire-order moments

For $y > 0$ and $\Re a > 0$ put

$$F_{a,s}(y) = y\Phi(-y, s, a), \quad \mathcal{E}_a(v) = v\zeta(v+1, a), \quad \mathcal{E}_a(0) = 1.$$

The last prescription removes the unique apparent pole. In particular, $F_{1,s}(y) = -\text{Li}_s(-y)$.

Theorem 1.1 (Entire-order reciprocal product). *For every $s, t \in \mathbb{C}$,*

$$\int_0^\infty F_{a,s}(y)F_{a,t}(1/y)\frac{dy}{y} = \mathcal{E}_a(s+t).$$

The integral is ordinary and absolutely convergent. It may be differentiated any finite number of times in either order.

Proof. With $f_{a,s}(x) = F_{a,s}(e^x)$, the Euler integral gives

$$f_{a,s}(x) = \frac{1}{\Gamma(s)} \int_0^\infty u^{s-1} e^{-(a-1)u} \frac{e^{x-u}}{1+e^{x-u}} du$$

for $\Re s > 0$. For $\max(0, 1 - \Re a) < c < 1$, its e^{-cx} -weighted version lies in $L^1 \cap L^\infty$. The same holds after any fixed derivative of the logistic kernel. The relation $(\partial_x + a - 1)f_{a,s} = f_{a,s-1}$ therefore extends it to an entire weighted family in s , with all logarithmic moments locally uniformly integrable.

For positive real parts of the two orders, the reciprocal y integral of their two Euler denominators equals $(u+v)/(1-e^{-(u+v)})$. Set $w = u+v$ and integrate the beta proportions. The result is

$$\frac{1}{\Gamma(s+t)} \int_0^\infty \frac{w^{s+t} e^{-(a-1)w}}{e^w - 1} dw = (s+t)\zeta(s+t+1, a).$$

Weighted holomorphy extends this identity to all orders and justifies all order derivatives. $\square$

Writing $L_m^a(y) = \partial_s^m F_{a,s}(y)|_{s=0}$ and using $\zeta(1+v, a) = v^{-1} + \sum_{n \geq 0} (-1)^n \gamma_n(a) v^n / n!$ gives

$$\int_0^\infty L_m^a(y) L_n^a(1/y) \frac{dy}{y} = \begin{cases} 1, & m+n=0, \\ (-1)^{m+n-1} (m+n) \gamma_{m+n-1}(a), & m+n > 0. \end{cases} \quad (1)$$

Thus this particular Stieltjes product law needs no finite-part normalization.

The full report gives a pole-free formula for every logarithmic moment. Let $W = s+t$, $Q_k(s, t) = ((s)_k - (-1)^k (t)_k) / (s+t+k-1)$, and $d_{2j} = [z^{2j}] \pi z / \sin \pi z$. The quotient Q_k is a polynomial: its numerator vanishes at $t = 1 - k - s$. Then

$$\int_0^\infty \log^\ell y F_{a,s}(y) F_{a,t}(1/y) \frac{dy}{y} = \ell! \sum_{j=0}^{\lfloor \ell/2 \rfloor} \frac{d_{2j}}{(\ell-2j+1)!} Q_{\ell-2j+1}(s, t) \mathcal{E}_a(W + \ell - 2j). \quad (2)$$

In particular the first moment is $(s - t)\mathcal{E}_a(W + 1)/2$, not twice that value. For a concise derivation, insert y^λ in the Euler-integral proof above. The result has the normally convergent local expansion

$$\frac{\pi}{\sin \pi \lambda} \sum_{k \geq 1} \frac{(s)_k - (-1)^k (t)_k}{k!} \zeta(W + k, a) \lambda^k.$$

Coefficient extraction and the polynomial divisibility give (2); entire continuation completes the proof.

At higher multiplicative depth, the convolution transform is $(\pi \csc \pi p)^r (p + a - 1)^{-W}$. The elementary polynomials $p_1 = 1$, $p_2 = X$, and $p_{r+2}(X) = (X^2 + r^2 \pi^2) p_r(X)$ give a finite central reduction. For example,

$$\mathcal{C}_{4,a}(W; 0) = \frac{(W)_2 \mathcal{E}_a(W + 2) + 4\pi^2 \mathcal{E}_a(W)}{6}.$$

Its value at $W = -2$ contains $1/3$ from the cancelled zeta pole. Every mixed logarithmic weight also has a finite differential-polynomial reduction, as proved in the full report.

Status boundary. This reciprocal geometry is distinct from the same-direction rational Mellin product. The arbitrary-order same-direction problem and the canonical Gaussian S_6/S_8 candidates are not resolved by this insertion. The ordinary formulas also do not replace the repository's separately normalized distributional Stieltjes products.